Singular Learning Theory

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Video: https://www.youtube.com/watch?v=SzVmiN8BvCc

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Thou shalt have the power to degenerate
—*On the Dignity of Man,* 1496

Introduction

Singular learning theory (SLT) is a theory of learning that accounts for *degeneracy* in neural networks. What is degeneracy? Let us first appeal to the Cambridge dictionary, documenting some features of typical uses of the word within mathematics:

> *Degenerate* (of an equation, curve, line, etc.) unusual or complicated in some way compared to other equations, curves, lines, etc. of a similar type, especially because a variable or parameter is zero.

On the other hand, we should recall that deep learning has as many roots in neuroscience as in mathematics. And as they say, "neural networks are *grown*." Perhaps we should also hear the biologist's definition:

> *Degenerate* (of an organism, chromosome, etc.) simpler than a form that previously existed because of no longer having a particular structure.

Actually, *both* of these definitions are apt. Within a typical neural network architecture, certain weight vectors correspond to neural networks from simpler architectures (often due to certain weights being zero). Structurally, these neural networks are simpler in form than their neighbours. Mathematically, they complicate the relationship between the neural network's parameter space and the resulting space of functions. In turn, learning in neural networks is substantially richer than learning in classical statistical models.

Singular learning theory is a framework for understanding learning that places degeneracy at the centre. The framework leverages powerful tools from algebraic geometry to understand, characterise, and resolve degeneracies, revealing their impact on learning. Unfortunately, the price of this mathematical power is that following the research literature without a deep background in pure mathematics is challenging.

That is where this tutorial comes in. We aim to distil the essence of singular learning theory—the notion of degeneracy and its role in learning—laying bare through simple examples and structured exercises the core intuitions driving research in the field. We believe understanding this essence requires no more than undergraduate-level mathematics. Moreover, to anyone armed with these core intuitions, understanding and contributing to research on the relationship between degeneracy and deep learning is within reach.

Contents. Specifically, the tutorial comprises the following four technical sections, each with definitions and guided exercises.

1. "Preliminaries" reviews core deep learning concepts, defining our notation and some running examples. The central concept is that of the parameter–function map.

2. "What is degeneracy?" defines degeneracy of parameter–function maps and relates it to symmetries, singularities of the Fisher information, and degenerate critical points of the loss landscape.

3. "The degeneracy hierarchy" derives the local learning coefficient (LLC) as a quantitative measure of the degree of degeneracy of a critical point via a volume scaling approach, and surveys several other perspectives on the quantity.

4. "Degeneracy and Bayesian deep learning" presents Watanabe's free energy formula connecting the LLC to Bayesian learning and discusses its implications.

We conclude with Section 5, providing pointers for further study and surveying the literature on SLT and deep learning.

Prerequisites. The following is an indicative list of mathematical concepts that will be helpful for reading the tutorial and completing the exercises.

- *Linear algebra:* vectors, matrices, rank, orthogonal matrices, rank–nullity, positive definiteness, eigenvalues, spectral decomposition.
- *Calculus:* partial derivatives, gradient, directional derivative, chain rule, Hessian, second-order Taylor expansion and remainder.
- *Integration and analysis:* multivariate integrals, change of variables, volume in $\mathbb{R}^d$, asymptotic notation (big-O, little-o), computing basic limits and integrals.
- *Probability:* probability simplex $\Delta(\cdot)$, conditional probability, probability density functions, independence, expectation, Bayes' rule, Gaussians, law of large numbers.

Section 1 reviews the basic framework of deep learning as parametric function approximation or statistical inference (parameter–function maps, deep linear networks, multi-layer perceptrons, loss functions, likelihood) along with Bayesian inference (prior, posterior, partition function, Bayesian free energy).

Readers with more advanced backgrounds may appreciate occasional references to topics from algebraic geometry, fractal geometry, or statistical physics. However, these references are tangential and readers without these backgrounds can safely skip them.

Fast-track. To paraphrase Euclid, there is no royal road to algebro-geometric learning theory. However, it is possible to get a bird's-eye view and the most important intuitions in a comparably short time. If this is what you are looking for, we recommend the following route through the tutorial. Assuming you are already somewhat comfortable with deep learning and Bayesian inference, skip Section 1, and refer back only as needed. Then, proceed as follows.

1. To understand parameter–function map versus loss landscape degeneracy: Read Section 2.1 and complete Exercises 2.1 and 2.2. Complete either Exercise 2.9 or Exercise 2.10. Read Section 2.5 and complete your choice of Exercise 2.14 and/or Exercise 2.15.

2. To understand the local learning coefficient via volume scaling: Read Section 3.1 and complete Exercises 3.1, 3.2, 3.4 and 3.7. Read Section 3.3 and Exercise 3.10.

3. To understand the relation between degeneracy and learning in the Bayesian case: Read all of Section 4 (it is shorter). Complete Exercise 4.2.

Once you are done, we hope you will consider taking the scenic route some other time.

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1 Preliminaries

In this section, we introduce basic terminology and notation for supervised deep learning and supervised Bayesian deep learning, along with some example neural network architectures and statistical models that we will repeatedly study throughout the tutorial.

1.1 Neural networks and parameter–function maps

Let $\mathcal{X}$ denote a space of network inputs (e.g., a space of images or token sequences encoded as vectors), and let $\mathcal{Y}$ denote a space of outputs (e.g., a space of numerical scores, class distributions, or next-token distributions).

Informally, a neural network architecture is a specification for how various **parameters** (e.g., neuron connection strengths / weights or neuron activation thresholds / biases) combine with input signals and each other to describe a function from $\mathcal{X}$ to $\mathcal{Y}$.

Formally, a neural network architecture essentially comprises a **parameter–function map** $\Phi : \mathcal{W} \to \mathcal{F}$, where $\mathcal{W} \subseteq \mathbb{R}^d$ is a d-dimensional **parameter space** and $\mathcal{F} \subseteq \mathcal{Y}^{\mathcal{X}}$ is some **hypothesis class** of functions from $\mathcal{X}$ to $\mathcal{Y}$. It is sometimes convenient to refer to the function $\Phi(w) : \mathcal{X} \to \mathcal{Y}$ as $f_w : \mathcal{X} \to \mathcal{Y}$.

Throughout the rest of this tutorial, we will study many different neural network architectures (parameter–function maps). The rest of this section explores some generally useful examples and some terminological points. To begin with, we have the following remark.

Remark (Parameters)

Note that the term "parameter" has two senses:

1. A "parameter" is an individual dimension of parameter space (e.g., "initialise this parameter to the value 0.0," or "this neural network has billions of parameters"). A loose synonym in this case is "weight."

2. A "parameter" is also a particular point in the parameter space (e.g., "this parameter is a local minimum of the loss function," or "the zero locus of this loss function contains a continuum of parameters"). A loose synonym in this case is "weight vector."

Both senses are in common usage in the literature as in this tutorial.

The simplest neural network architecture models a single "neuron" with d inputs $x_1, \ldots x_d$. Each input x_i is multiplied by an incoming weight w_i to produce the neuron's output. We formalise this case in Example 1.1.

Example 1.1 (A linear neuron)**.** Let $\mathcal{X} = \mathbb{R}^d$ for some positive integer d and let $\mathcal{Y} = \mathbb{R}$. Define a parameter space $\mathcal{W} = \mathbb{R}^d$. Define a parameter–function map that maps a column vector $w \in \mathcal{W}$ to the function $f_w : \mathbb{R}^d \to \mathbb{R}$ such that for $x \in \mathbb{R}^d$,

$$f_w(x) = w^\top x = \sum_{i=1}^{d} w_i x_i. \tag{1}$$

Let us generalise this basic neural network in three ways: to add multiple outputs, "depth," and non-linearity. First, we can generalise from a scalar output to a vector output as follows.

Example 1.2 (Multi-linear neural network)**.** Let $\mathcal{X} = \mathcal{Y} = \mathbb{R}^m$ for some positive integer m. Define a parameter space $\mathcal{W} = \mathbb{R}^{m^2}$. Let each parameter $w \in \mathcal{W}$ encode an $m \times m$ matrix $W \in \mathbb{R}^{m \times m}$. The parameter–function map transforms each parameter $w \in \mathcal{W}$ to a function $f_w : \mathbb{R}^m \to \mathbb{R}^m$ such that for $x \in \mathbb{R}^m$,

$$f_w(x) = Wx. \tag{2}$$

If we take each row of W to encode the weight vector of a linear neuron (Example 1.1), we see that we have produced m outputs by stacking m independent linear neurons together. Such a group of neurons is called a **layer**.

Remark (Encodings)

In Example 1.2, the parameter space is formally $\mathcal{W} = \mathbb{R}^{m^2}$, but we find it convenient to identify each parameter vector $w \in \mathbb{R}^{m^2}$ with the $m \times m$ matrix W it encodes. We would thus write f_W in place of f_w. More generally, whenever the parameters of an architecture naturally decompose into named matrices or vectors, we index the parameter–function map by these structured objects directly rather than by the underlying vector w. The remaining examples in this section demonstrate this convention, and we continue to use it throughout this tutorial.

Next, let's add *depth* to our neural network by composing two layers together.

Example 1.3 (Deep linear network; DLN)**.** Let $\mathcal{X} = \mathcal{Y} = \mathbb{R}^m$ for some positive integer m. Let h be a positive integer and define the parameter space $\mathcal{W} = \mathbb{R}^{2mh}$, with each parameter encoding a pair of matrices $A \in \mathbb{R}^{h \times m}$ and $B \in \mathbb{R}^{m \times h}$. The parameter–function map sends (A, B) to $f_{A,B} : \mathbb{R}^m \to \mathbb{R}^m$ such that for $x \in \mathbb{R}^m$,

$$f_{A,B}(x) = BAx. \tag{3}$$

The above architecture is called a two-layer **deep linear network (DLN)**. The architecture can be interpreted as composing two multi-linear neural networks. The vector

of outputs of the neurons of the first network becomes the vector of inputs to the second network. The intermediate output vectors are called **activations**.

Note that if $h \geq m$, the two-layer DLN architecture indexes the same hypothesis class as in Example 1.2, that of linear transforms on $\mathbb{R}^m$. If $h < m$, the hypothesis class includes only transforms with rank up to h.

Finally, we'll add *non-linearity* between pairs of layers. To do so, we introduce a non-linear scalar function $\sigma : \mathbb{R} \to \mathbb{R}$, called an *activation function*, to transform the output of each intermediate neuron. Common examples of activation functions, which we will study in this tutorial, include the following:

1. The **hyperbolic tangent** function $\tanh(z) = \dfrac{e^z - e^{-z}}{e^z + e^{-z}}$.
2. The **rectified linear unit (ReLU)** function $\text{relu}(z) = \max(z, 0)$.

This results in the following non-linear neural network architecture.

Example 1.4 (Multi-layer perceptron; MLP)**.** Define input, output, and parameter spaces as in Example 1.3. Let $\sigma : \mathbb{R} \to \mathbb{R}$ be an activation function. The parameter–function map sends (A, B) to the function $f_{A,B} : \mathbb{R}^m \to \mathbb{R}^m$ such that for $x \in \mathbb{R}^m$,

$$f_{A,B}(x) = B \cdot \sigma(Ax), \tag{4}$$

where we lift the activation function to operate element-wise over column vectors $Ax \in \mathbb{R}^h$.

This kind of architecture is called a **multi-layer perceptron (MLP).** Note that the two-layer DLN is recovered if we use the identity function as an activation function. However, if we use a non-linear activation function, we can index a much richer hypothesis class.

The expressivity of these architectures is limited, however, by the omission of a basic detail—the inclusion of a bias parameter for each neuron. We invite the reader to correct this omission as our first exercise, which serves as a chance to familiarise oneself with the concept of a parameter–function map.

Exercise 1.1 (Biased neurons). A biased neuron is a neuron with an additional parameter that is added to the weighted sum of its inputs before it produces its output. Bias parameters allow each layer to represent an *affine* transform, rather than just a linear transform. For each of Examples 1.1 to 1.4, extend the example to use biased neurons. Precisely define the parameter space and the parameter–function map in each case.

Modern deep learning leverages more involved parameter–function maps. Neural network architectures are typically defined in a modular fashion, including linear (or affine)

layers and non-linear layers like the above, along with more specialised layers (well-known examples including convolutional layers, residual layers, and attention layers).

1.2 Supervised deep learning and loss functions

In a typical supervised deep learning setting, we are given a data set of pairs $(x^{(1)}, y^{(1)})$, $\ldots$, $(x^{(n)}, y^{(n)}) \in \mathcal{X} \times \mathcal{Y}$. These pairs exemplify inputs and their corresponding (possibly noisy) outputs from an unknown target function $f : \mathcal{X} \to \mathcal{Y}$ which we would like to approximate using a neural network. That is, we would like to find a parameter $w \in \mathcal{W}$ such that the corresponding function f_w approximates the target function f.

Formally, given an example $(x, y) \in \mathcal{X} \times \mathcal{Y}$, define a **per-example loss** function $L_{x,y} : \mathcal{W} \to \mathbb{R}$ to measure the deviation of $f_w(x)$ from the output y. A typical loss function for vector outputs $\mathcal{Y} \subset \mathbb{R}^m$ is to use the **squared error** loss

$$L_{x,y}(w) = \|f_w(x) - y\|^2. \tag{5}$$

If $\mathcal{Y} = \Delta^{m-1}$ is a space of discrete distributions over m objects, it is typical to use **cross entropy** loss

$$L_{x,y}(w) = -\sum_{i=1}^{m} y(i) \log f_w(i \mid x). \tag{6}$$

Given a per-example loss function and a **data distribution** $q \in \Delta(\mathcal{X} \times \mathcal{Y})$ over the space of examples (representing our possibly-noisy target function), we formalise the objective of supervised learning as finding $w \in \mathcal{W}$ so as to minimise the **population loss** function $L : \mathcal{W} \to \mathbb{R}$ defined such that

$$L(w) = \mathbb{E}_{(x,y)\sim q}[L_{x,y}(w)]. \tag{7}$$

The population loss aggregates differences on outputs for individual inputs into an overall measure of difference between f_w and the target function.

In practice, we often can't evaluate the population loss over the entire input/output space. We instead optimise an estimator based on our data set of n input–output pairs assumed to be sampled independently and identically from q. Define an **empirical loss** function $L_n : \mathcal{W} \to \mathbb{R}$ such that

$$L_n(w) = \frac{1}{n} \sum_{i=1}^{n} L_{x^{(i)},y^{(i)}}(w). \tag{8}$$

If the per-example loss is squared error, the empirical loss is known as the **mean squared error** objective.

$$L_n(w) = \frac{1}{n} \sum_{i=1}^{n} \left\| f_w(x^{(i)}) - y^{(i)} \right\|^2. \tag{9}$$

Exercise 1.2 (Empirical loss and population loss). Fix $w \in \mathcal{W}$. Prove the following properties of the relationship between the empirical loss $L_n(w)$ and the population loss $L(w)$.

(a) For any n, the empirical loss is an unbiased estimator of the population loss. That is,

$$\mathbb{E}_{(x^{(i)},y^{(i)})\sim q}[L_n(w)] - L(w) = 0. \tag{10}$$

(b) As $n \to \infty$, the empirical loss converges almost surely to the population loss.

Given a loss function, a **training algorithm** is a search algorithm that aims to find $w \in \mathcal{W}$ such that the loss is approximately minimised. Most modern deep learning algorithms are variants of **stochastic gradient descent (SGD)**, which implements an iterative gradient-based local search of the parameter space using empirical loss on subsamples of the data set.

Through many impressive feats of computer science and hardware/software engineering, we are able to run such training algorithms to find low-loss parameters within parameter spaces with billions of dimensions. The details are vitally important to the success of modern deep learning, but are beyond the scope of this tutorial. We only mention SGD in order to emphasise that *any local search method depends intimately on the properties of the parameter–function map.*

1.3 Statistical models and parameter–distribution maps

The previous sections introduce neural networks and supervised deep learning within a framework of function approximation. An alternative approach is to view supervised learning as statistical inference, and particularly Bayesian inference. This statistical framework is commonly used within the SLT literature, and we will encounter it within this tutorial.

The first step to upgrading our framework is to move from talking about functions to talking about conditional distributions. For each neural network, we should be able to determine not just the output corresponding to each input, but a probability density measuring how likely each output is given each input.

Formally, let $\mathcal{D} \subseteq \Delta(\mathcal{Y})^{\mathcal{X}}$ represent a class of conditional distributions over $\mathcal{Y}$. A distributional neural network architecture comprises a **parameter–distribution map** $\Psi : \mathcal{W} \to \mathcal{D}$. It is sometimes convenient to refer to the conditional distribution $\Psi(w) : \mathcal{X} \to \Delta(\mathcal{Y})$ using the notation $p_w : \mathcal{X} \to \Delta(\mathcal{Y})$. The probability mass/density of a given output $y \in \mathcal{Y}$ conditional on a given input $x \in \mathcal{X}$ is typically denoted either $p_w(y \mid x)$ or $p(y \mid x, w)$.

Sometimes, defining a parameter–distribution map is simply a re-framing. For example, in image classification, our output space $\mathcal{Y}$ was a space of distributions over image

classes. Let $\mathcal{C}$ represent the underlying class space, then $\mathcal{Y} = \Delta(\mathcal{C})$ and we can view our parameter–function map $\Phi : \mathcal{W} \to \mathcal{F}$ as a parameter–distribution map $\Psi : \mathcal{W} \to \mathcal{D}$ with $\mathcal{D} = \mathcal{F} \subseteq \mathcal{Y}^{\mathcal{X}} = \Delta(\mathcal{C})^{\mathcal{X}}$.

In other cases, we can easily construct a parameter–distribution map from a parameter–function map by introducing a **noise model**. For example, in regression with an output space $\mathcal{Y} = \mathbb{R}^m$, given a parameter–function map $w \mapsto f_w$ we can define a parameter–distribution map $\Psi : \mathcal{W} \to \Delta(\mathcal{Y})^{\mathcal{X}}$ such that, for $w \in \mathcal{W}$ and $x \in \mathcal{X}$,

$$p(y \mid x, w) = \frac{1}{(2\pi)^{m/2}} \exp\left(-\frac{\|y - f_w(x)\|^2}{2}\right). \tag{11}$$

The above **Gaussian noise model** amounts to modelling each output y as drawn given x, w according to the rule $y = f_w(x) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I_m)$.

As the following exercises explore, performing statistical inference according to the principle of maximum likelihood aligns with the loss-minimisation perspective (given an appropriate choice of loss function).

Exercise 1.3 (Maximum likelihood as MSE minimisation). Consider a regression problem with $\mathcal{X} = \mathcal{Y} = \mathbb{R}^m$. Assume we have a neural network architecture with a parameter space $\mathcal{W}$. Form a corresponding statistical model using the Gaussian noise model of (11). Let $(x^{(1)}, y^{(1)}), \ldots, (x^{(n)}, y^{(n)}) \in \mathcal{X} \times \mathcal{Y}$ be a data set of n regression examples.

(a) Write an expression for the conditional probability of observing the labels $Y_n = (y^{(1)}, \ldots, y^{(n)})$ given the inputs $X_n = (x^{(1)}, \ldots, x^{(n)})$ and some fixed parameter $w \in W$, assuming the labels are sampled independently.

(b) Denote the resulting quantity by $p(Y_n \mid X_n, w)$. It is called the **likelihood** of the data set according to the model w. Compute and simplify the **negative log-likelihood**, given by $-\log p(Y_n \mid X_n, w)$.

(c) Show that if L_n is the mean squared error loss function from (9), then

$$\arg\max_{w \in \mathcal{W}} p(Y_n \mid X_n, w) = \arg\min_{w \in \mathcal{W}} L_n(w).$$

Exercise 1.4 (Maximum likelihood as cross entropy minimisation). Consider a classification problem with $\mathcal{Y} = \Delta^{m-1}$. Assume we have a neural network architecture with a parameter space $\mathcal{W}$. As discussed, the parameter–function map in this case corresponds to a parameter–distribution map to conditional distributions over the underlying discrete class space $\mathcal{C} = \{1, \ldots, m\}$. Let $(x^{(1)}, c^{(1)}), \ldots, (x^{(n)}, c^{(n)}) \in \mathcal{X} \times \mathcal{C}$ be a data set of n labelled classification examples. For each $c^{(i)} \in \mathcal{C}$ define

$$y^{(i)} = \delta_{c^{(i)}} \in \Delta^{m-1} \tag{12}$$

to be the corresponding Dirac distribution (with probability mass 1 concentrated on $c^{(i)}$).

(a) Write an expression for the likelihood of the data set, $p(C_n \mid X_n, w)$, that is, the conditional probability of observing the labels $C_n = (c^{(1)}, \ldots, c^{(n)})$ given the inputs $X_n = (x^{(1)}, \ldots, x^{(n)})$ and some fixed parameter $w \in W$. Assume the labels are sampled independently.

(b) Show that the negative log-likelihood

$$-\log p(C_n \mid X_n, w) = -\sum_{i=1}^{n} \log f_w(c^{(i)} \mid x^{(i)}).$$

(c) Using (12), show that if the per-example loss function for the optimisation problem is cross entropy loss (6), then

$$\arg\max_{w \in \mathcal{W}} p(C_n \mid X_n, w) = \arg\min_{w \in \mathcal{W}} L_n(w).$$

Exercise 1.5 (Maximum likelihood as loss minimisation, in general). Consider a neural network architecture with a parameter space $\mathcal{W}$, parameter–function map Φ, and output space $\mathcal{Y} = \mathbb{R}^m$. Let $L_{x,y} : \mathcal{W} \to \mathbb{R}$ be a per-example loss function. Assume that for each $x \in \mathcal{X}$ and $w \in \mathcal{W}$, the function $y \mapsto \exp(-L_{x,y}(w))$ is integrable over $\mathcal{Y}$. Define a corresponding statistical model with parameter–distribution map Ψ mapping $w \in \mathcal{W}$ to $p_w : \mathcal{X} \to \Delta(\mathcal{Y})$ such that for all $y \in \mathcal{Y}$, $x \in \mathcal{X}$, and $w \in \mathcal{W}$, we have

$$p(y \mid x, w) = \frac{\exp(-L_{x,y}(w))}{Z(x, w)}, \qquad Z(x, w) = \int_{\mathcal{Y}} \exp(-L_{x,z}(w))\, dz.$$

We re-use the notation X_n, Y_n from Exercise 1.3, and assume that labels are sampled independently.

(a) Show that the negative log-likelihood decomposes as

$$-\log p(Y_n \mid X_n, w) = \sum_{i=1}^{n} L_{x^{(i)}, y^{(i)}}(w) + \sum_{i=1}^{n} \log Z(x^{(i)}, w).$$

(b) Now suppose the loss function takes the form $L_{x,y}(w) = \ell(f_w(x) - y)$ for some integrable function $\ell : \mathbb{R}^m \to \mathbb{R}$. Show that $Z(x, w)$ is independent of w, and conclude that

$$\arg\max_{w \in \mathcal{W}} p(Y_n \mid X_n, w) = \arg\min_{w \in \mathcal{W}} L_n(w).$$

Remark (Negative log-likelihood loss)

Exercises 1.3 to 1.5 nominally show that some maximum likelihood estimation problems have the same sets of ideal solutions as some carefully-chosen optimisation problems ($\arg\min = \arg\max$). In general, that the global optima for two optimisation problems coincide is insufficient to show that practical optimisation methods will always find the same solutions: practical optimisation methods don't always find global optima. However, in the above cases, your derivation likely reveals a stronger result: the optimisation landscapes for the function approximation problem are related to the optimisation landscape for the maximum likelihood estimation problem by a strictly monotonically decreasing transform (an affine transform of a logarithm).

1.4 Bayesian deep learning

The principle of maximum likelihood, explored in the preceding exercises, is one approach to statistical inference. An alternative approach is **Bayesian inference**, which treats w as a random variable and maintains a probability distribution over $\mathcal{W}$ reflecting our uncertainty about which parameter best explains the data. The majority of theoretical results in SLT have been developed within this setting (Watanabe, 2009, 2018, cf.,).

Bayesian learning begins with a **prior** distribution $\varphi \in \Delta(\mathcal{W})$, a probability density on $\mathcal{W}$ representing our beliefs about the parameter w before observing any data. Throughout this tutorial, we assume φ is positive and smooth on $\mathcal{W}$.

Definition 1.5 (Posterior distribution, partition function, free energy)**.** Given a data set of n input–output pairs $(x^{(1)}, y^{(1)}), \ldots, (x^{(n)}, y^{(n)}) \in \mathcal{X} \times \mathcal{Y}$, Bayes' rule updates the prior into the **posterior distribution** $\pi_n \in \Delta(\mathcal{W})$, a probability density on $\mathcal{W}$ defined by

$$\pi_n(w) = \frac{1}{Z_n}\varphi(w)\prod_{i=1}^{n} p(y^{(i)} \mid x^{(i)}, w) \tag{13}$$

where $Z_n \in \mathbb{R}$ is a normalising constant called the **partition function** (or **marginal likelihood** or **model evidence**),

$$Z_n = \int_{\mathcal{W}} \varphi(w)\prod_{i=1}^{n} p(y^{(i)} \mid x^{(i)}, w)\, dw. \tag{14}$$

We often work with the **(Bayesian) free energy**, the negative log of the model evidence,

$$F_n = -\log Z_n. \tag{15}$$

The posterior re-weights the prior by the likelihood of the observed data: parameters w under which the data is more probable receive higher posterior density, while parameters under which the data is improbable are down-weighted.

The partition function Z_n is the marginal probability of observing the data set, averaged over all parameters weighted by the prior. It quantifies how well the statistical model *as a whole* predicts the observed data. The free energy is an inverted measure of how well the model fits the data.

Exercise 1.6 (Bayesian posterior as a Gibbs distribution). To interpret Bayesian inference in terms more similar to function approximation, we introduce a loss function based on the **negative log-likelihood**,

$$L_{x,y}(w) = -\log p(y \mid x, w), \tag{16}$$

$$L_n(w) = -\frac{1}{n}\sum_{i=1}^{n} \log p(y^{(i)} \mid x^{(i)}, w). \tag{17}$$

Here, $p(y \mid x, w)$ is the conditional density from the parameter–distribution map and $(x^{(1)}, y^{(1)}), \ldots, (x^{(n)}, y^{(n)}) \in \mathcal{X} \times \mathcal{Y}$ is a data set of n examples. Show that

$$\pi_n(w) = \frac{1}{Z_n}\varphi(w)\exp\bigl(-nL_n(w)\bigr)$$

and

$$Z_n = \int_{\mathcal{W}} \varphi(w)\exp\bigl(-nL_n(w)\bigr)\, dw.$$

The **Bayesian deep learning process** is the process of re-weighting each individual parameter vector in response to seeing an increasing amount of data, producing the sequence of belief distributions $\varphi, \pi_1, \pi_2, \ldots \in \Delta(\mathcal{W})$. Over time, the posterior will concentrate around parameters which provide a good fit for the data (though there is more to the story in the singular case, as we will see in later sections).

Bayesian deep learning is a *global search* in that all parameter vectors are considered in parallel. This makes exact Bayesian learning computationally intractable for non-trivial models, but more analytically tractable than SGD in general. Moreover, as we will explore in later sections, the dynamics of Bayesian learning still reflect the local geometry of the parameter–distribution map.

In preparation for exploring this connection between geometry and learning, we finally introduce localised variants of the partition function and free energy, where the integral in (14) is restricted to a certain neighbourhood in parameter space.

Definition 1.6 (Local partition function and local free energy)**.** Given a neigh-

bourhood $\mathcal{U} \subseteq \mathcal{W}$, the **local partition function** and **local free energy** are

$$Z_n(\mathcal{U}) = \int_{\mathcal{U}} \varphi(w) \prod_{i=1}^{n} p(y^{(i)} \mid x^{(i)}, w)\, dw, \qquad F_n(\mathcal{U}) = -\log Z_n(\mathcal{U}). \tag{18}$$

The local partition function (free energy) measures how well (poorly) parameters within a given neighbourhood collectively explain the data. Note, $Z_n(\mathcal{W}) = Z_n$ and $F_n(\mathcal{W}) = F_n$.

Exercise 1.7 (Local posterior mass and local free energy). Let $\mathcal{U}, \mathcal{V} \subseteq \mathcal{W}$ be two neighbourhoods of parameter space. Let $\pi_n(\mathcal{U}) = \int_{\mathcal{U}} \pi_n(w)\, dw$ denote the posterior mass of a neighbourhood. Show the following.

(a) $\pi_n(\mathcal{U}) = \dfrac{Z_n(\mathcal{U})}{Z_n}$ (the same holds for $\mathcal{V}$).

(b) $\log \dfrac{\pi_n(\mathcal{U})}{\pi_n(\mathcal{V})} = F_n(\mathcal{V}) - F_n(\mathcal{U})$.

In conclusion, the posterior odds ratio of the two regions $\pi_n(\mathcal{U})/\pi_n(\mathcal{V})$ depends exponentially on the difference between their local free energies.

Exercise 1.8 (Partition function of partitioned parameter space). Let $\mathcal{U}_1, \ldots, \mathcal{U}_k \subseteq \mathcal{W}$ be disjoint subsets with $\mathcal{W} = \mathcal{U}_1 \sqcup \cdots \sqcup \mathcal{U}_k$.

(a) Show that the partition function decomposes as $Z_n = \sum_{j=1}^{k} Z_n(\mathcal{U}_j)$, and hence

$$F_n = -\log \sum_{j=1}^{k} e^{-F_n(\mathcal{U}_j)}.$$

(b) Show that if $F_n(\mathcal{U}_1) < F_n(\mathcal{U}_j)$ for all $j \neq 1$, then

$$F_n(\mathcal{U}_1) - \log k < F_n < F_n(\mathcal{U}_1).$$

In conclusion, the overall free energy is dominated by the region(s) with the lowest local free energy.

2 What is degeneracy?

In this section, we define degeneracy as a property of parameter–function maps, and study the relationship between this property and other similar notions.

2.1 A definition of degeneracy

Consider a neural network architecture with parameter space $\mathcal{W} \subseteq \mathbb{R}^d$ and parameter–function map $\Phi : \mathcal{W} \to \mathcal{F}$ $(w \mapsto f_w)$. Say the parameter–function map is **degenerate at** $w \in \mathcal{W}$ if there is a non-zero vector $v \in \mathbb{R}^d$ such that the directional derivative of the parameter–function map in the direction v is zero:

$$D_v\Phi(w) = 0. \tag{19}$$

The directional derivative $D_v\Phi(w)$ for non-zero $v \in \mathbb{R}^d$ may be defined variously as

$$D_v\Phi(w) = \frac{v}{\|v\|} \cdot \nabla\Phi(w) = \sum_{i=1}^{d} \frac{v_i}{\|v\|} \frac{\partial\Phi}{\partial w_i}(w) = \lim_{\epsilon \to 0} \frac{\Phi(w + \epsilon v) - \Phi(w)}{\epsilon\|v\|}. \tag{20}$$

We conventionally define $D_0\Phi(w) = 0$. Note that w is fixed, but the directional derivative is still a function (of the same type as $\Phi(w)$; $D_v\Phi(w) : \mathcal{X} \to \mathcal{Y}$), since it represents an infinitesimal change in $\Phi(w)$ in response to a change in w in the direction v. In (19) we mean that this function is identically zero for all inputs $x \in \mathcal{X}$ for the given $w \in \mathcal{W}$.

Intuitively, the parameter–function map is degenerate at $w \in \mathcal{W}$ if there is a direction in which we can infinitesimally perturb w without changing the function f_w.

This definition of degeneracy applies to a single point in parameter space. As we shall soon see, it may be the case that the parameter–function map is degenerate at some points but not at others. We can clarify the situation with new terminology.

- We say that a parameter–function map is **somewhere degenerate** if it is degenerate at *any* point in parameter space.
- We say that a parameter–function map is **everywhere degenerate** if it is degenerate at *all* points in parameter space.

By convention, if we say that the parameter–function map is simply **degenerate** (without specifying somewhere, everywhere, or at a particular point), we mean that it is *somewhere degenerate*.

The following exercises explore this definition in some toy parametrisations of a simple function class (constants), displaying in the simplest possible setting some basic forms of degeneracy that will arise repeatedly throughout the tutorial.

Exercise 2.1 (Parametrising the space of constants). Let $\mathcal{X} = \{*\}$ and $\mathcal{Y} = \mathbb{R}$, so that we have a hypothesis class of constants. Consider the scalar parameter space $\mathcal{W} = \mathbb{R}$ and the parameter–function map that maps w to $f_w = w$ (the output is just the parameter itself).

(a) What is the directional derivative of the parameter–function map in direction $v = 1$?

Hint: What kind of derivative does this reduce to?

(b) At which points in the parameter space is this parameter–function map degenerate, if any?

Exercise 2.2 (Degeneracy from raising a parameter to a power). Again, consider a hypothesis class of constants and the scalar parameter space $\mathcal{W} = \mathbb{R}$. This time, define a parameter–function map that maps w to $f_w = w^3$.

(a) Show that this architecture indexes exactly the same hypothesis class as the architecture described in Exercise 2.1.

(b) What is the directional derivative of the parameter–function map in direction $v = 1$?

Hint: What kind of derivative does this reduce to?

(c) At which points in the parameter space is the parameter–function map degenerate, if any?

Exercise 2.3 (Degeneracy from multiplying two parameters together). Again, consider a hypothesis class of constants. Consider this time the two-dimensional parameter space $\mathcal{W} = \mathbb{R}^2$. Define a parameter–function map that maps $w = (a, b)$ to $f_w = a \cdot b$.

(a) Show that this architecture indexes exactly the same hypothesis class as the architecture described in Exercise 2.1.

(b) What is the directional derivative of the parameter–function map in direction $v = (1, 0)$?

Hint: What kind of derivative does this reduce to?

(c) What is the directional derivative of the parameter–function map in direction $v = (0, 1)$?

Hint: What kind of derivative does this reduce to?

(d) At which points in the parameter space is the parameter–function map degenerate *in these directions,* if any?

(e) At which points in the parameter space is the parameter–function map degenerate, if any?

Exercise 2.4 (Degeneracy from adding two parameters together). Again, consider a hypothesis class of constants. Consider this time the two-dimensional parameter space $\mathcal{W} = \mathbb{R}^2$. Define a parameter–function map that maps $w = (a, b)$ to $f_w = a + b$.

(a) Show that this architecture indexes exactly the same hypothesis class as the architecture described in Exercise 2.1.

(b) What is the directional derivative of the parameter–function map in direction $v = (1, -1)$?

(c) At which points in the parameter space is the parameter–function map degenerate, if any?

2.2 Degeneracy and continuous symmetries

A common source of degeneracy in neural network architectures is the presence of continuous symmetries of the parameter–function map. A continuous symmetry traces out a curve of functionally equivalent parameters. The tangent to this curve is a degenerate direction. In this section, we will explore this kind of symmetry and some examples from deep learning.

A **symmetry** of a parameter–function map Φ is a transformation of parameter space that maps parameters while preserving the implemented function. That is, a transformation $T : \mathcal{W} \to \mathcal{W}$ is a symmetry if for all $w \in \mathcal{W}$, we have $\Phi(w) = \Phi(T(w))$ ($f_w = f_{T(w)}$).

A **continuous symmetry** of a parameter–function map Φ is a family of transformations $\{T_t : \mathcal{W} \to \mathcal{W}\}_{t \in \mathbb{R}}$ indexed by a parameter $t \in \mathbb{R}$, such that

(i) T_t is a symmetry ($\Phi \circ T_t = \Phi$) for all $t \in \mathbb{R}$,

(ii) T_0 is the identity transformation on $\mathcal{W}$, and

(iii) The map $t \mapsto T_t(w)$ is differentiable for each $w \in \mathcal{W}$.

The continuous symmetry is **trivial at** w if $\frac{d}{dt}\big|_{t=0} T_t(w) = 0$, or else **non-trivial at** w.

Exercise 2.5 (Continuous symmetry example). Recall the parameter–function map from Exercise 2.4, with $\mathcal{W} = \mathbb{R}^2$ and with $(a, b) \in \mathcal{W}$ mapping to the constant function $f_{a,b} = a + b$. Consider the family of transformations $T_t : \mathcal{W} \to \mathcal{W}$ for $t \in \mathbb{R}$ such that

$$T_t(a, b) = (a + t, b - t).$$

(a) Describe the effect of this family of transformations on the parameter space.

(b) Show that this family of transformations is a continuous symmetry.

(c) Show that this continuous symmetry is non-trivial everywhere.

At each parameter, a non-trivial continuous symmetry traces out a curve of functionally equivalent parameters. This indicates the presence of a direction in parameter space in which the function does not change. The following proposition formalises this connection between continuous symmetries and degeneracy.

Proposition 2.1. *If a parameter–function map Φ admits a continuous symmetry $\{T_t\}$ that is non-trivial at w, then Φ is degenerate at w.*

Proof. Define the curve in parameter space traced by the continuous symmetry, $\gamma : \mathbb{R} \to \mathcal{W}$ with $\gamma(t) = T_t(w)$. Fix $t_0 \in \mathbb{R}$. Since T_t is a continuous symmetry, the derivative along the curve at t_0 vanishes:

$$\begin{aligned} D_{\gamma'(t_0)}\Phi(\gamma(t_0)) &\propto \gamma'(t_0) \cdot \nabla\Phi(\gamma(t_0)) \\ &= \left.\frac{d}{dt}\right|_{t=t_0} \Phi(\gamma(t)) && \text{(chain rule)} \\ &= \left.\frac{d}{dt}\right|_{t=t_0} \Phi(w) && \text{(symmetry, } \Phi \circ T_t = \Phi) \\ &= 0. && (\Phi(w) \text{ is constant in } t) \end{aligned}$$

In particular, at $t = 0$, we have $0 = D_{\gamma'(0)}\Phi(\gamma(0)) = D_{\gamma'(0)}\Phi(w)$ since $\gamma(0) = w$ by identity. Since T_t is non-trivial at w, $\gamma'(0)$ is non-zero, so $\gamma'(0)$ is a degenerate direction at w. □

Corollary 2.2. *If a parameter–function map Φ admits a continuous symmetry $\{T_t\}$ that is non-trivial at every parameter $w \in \mathcal{W}$, then Φ is everywhere degenerate.*

The following exercises exhibit some continuous symmetries in two neural network architectures, a small ReLU MLP and a toy autoencoder. Similar architectures are studied in more detail from an SLT perspective by Carroll (2021) and Chen et al. (2023) respectively.

Exercise 2.6 (ReLU scaling symmetry). Consider a two-layer MLP (Example 1.4) with $\sigma = \mathrm{relu}$, scalar inputs and outputs ($m = 1$), and a single hidden unit ($h = 1$). The parameter space is $\mathcal{W} = \mathbb{R}^2$ with parameters (a, b), and the parameter–function map is $f_{a,b}(x) = b \cdot \mathrm{relu}(ax)$.

(a) Show that relu is *positively homogeneous*: $\mathrm{relu}(\alpha z) = \alpha\,\mathrm{relu}(z)$ for all $\alpha > 0$ and $z \in \mathbb{R}$.

(b) Define $T_t(a, b) = (e^t a,\, e^{-t} b)$ for $t \in \mathbb{R}$. Show that $\{T_t\}$ is a continuous symmetry of this parameter–function map.

(c) For a fixed parameter $(a, b) \neq (0, 0)$, compute the degenerate direction arising from this symmetry at (a, b).

(d) Show that the symmetry is trivial at the origin. Is this parameter–function map degenerate at the origin?

Exercise 2.7 (Rotation symmetry in a linear autoencoder). Consider a *linear autoencoder* with bottleneck dimension h and ambient dimension $m \geq h$. The parameter space encodes a single matrix $W \in \mathbb{R}^{h \times m}$ and the parameter–function map sends W to the function $\Phi(W) = f_W : \mathbb{R}^m \to \mathbb{R}^m$ where

$$f_W(x) = W^\top W x$$

for $x \in \mathbb{R}^m$. Here, W serves simultaneously as encoder ($x \mapsto Wx \in \mathbb{R}^h$) and decoder ($z \mapsto W^\top z \in \mathbb{R}^m$).

(a) Show that for any orthogonal matrix $R \in \mathbb{R}^{h \times h}$ (i.e., any $h \times h$ matrix such that $R^\top R = I$), the map $T_R(W) = RW$ is a symmetry.

(b) Specialise to $h = 2$. Using the rotation matrices

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

show that $T_\theta(W) = R(\theta)W$ defines a continuous symmetry.

(c) For a fixed parameter $W \in \mathbb{R}^{2 \times m}$, compute the degenerate direction that arises from this symmetry.

(d) For general bottleneck dimension h, how many independent continuous symmetries does the orthogonal group $O(h)$ contribute?

Hint: What is the dimension of $O(h)$?

2.3 Localised degeneracy

We have seen that a non-trivial continuous symmetry traces out curves of functionally equivalent parameters throughout the entire parameter space, and the tangent directions to these curves are degenerate directions at every point. However, not all degenerate directions arise from globally continuous symmetries. In fact, the more interesting case is when degeneracy affects only some parameters and not others.

Within certain subsets of parameter space, many neural network architectures (including those studied in Exercises 2.6 and 2.7) exhibit degenerate directions that cannot be defined in terms of continuous transformations that are globally symmetries. In this section, we explore localised degeneracies of this kind.

We begin with some examples of transformations that are only continuous symmetries within certain subsets of parameter space.

Exercise 2.8 (Localised symmetry example). Recall the parameter–function map from Exercise 2.3, with $\mathcal{W} = \mathbb{R}^2$ and with $(a, b) \in \mathcal{W}$ mapping to the constant function $f_{a,b} = a \cdot b$. Consider the two families of transformations $T_t^{(a)}, T_t^{(b)} : \mathcal{W} \to \mathcal{W}$ for $t \in \mathbb{R}$ such that

$$T_t^{(a)}(a, b) = (a + t, b), \qquad T_t^{(b)}(a, b) = (a, b + t).$$

(a) Describe the effect of each family of transformations on the parameter space.

(b) In which subset of parameter space does each family of transformations constitute a symmetry for all $t \in \mathbb{R}$?

(c) Compare your results to your answer to Exercise 2.3(d).

The above example is technically a two-layer DLN with $m = h = 1$. Let us now explore localised degeneracy in a general two-layer DLN and then in a non-trivial MLP.

Exercise 2.9 (Degeneracy in the two-layer DLN). Consider the two-layer DLN from Example 1.3, with $h = m$. The parameter space encodes two matrices $A, B \in \mathbb{R}^{m \times m}$, and the parameter–function map sends (A, B) to the function $\Phi(A, B) = f_{A,B} : \mathbb{R}^m \to \mathbb{R}^m$ such that $f_{A,B}(x) = BAx$ for $x \in \mathbb{R}^m$.

(a) Let $(\delta A, \delta B)$ be a unit perturbation in parameter space (a unit vector in the underlying parameter space $\mathbb{R}^{2m^2}$, decoded into a pair of matrices). Show that the directional derivative of the parameter–function map at (A, B) in direction $(\delta A, \delta B)$ is the linear map

$$D_{(\delta A, \delta B)}\Phi(A, B) = B\,\delta A + \delta B\,A.$$

That is, $D_{(\delta A, \delta B)}\Phi(A, B)x = (B\,\delta A + \delta B\,A)x$.

Hint: Use the limit definition, (20).

(b) Show that at the zero parameter $(A, B) = (0, 0)$, every direction in parameter space is degenerate. Count the number of dimensions in the subspace of degenerate directions.

(c) Suppose both A and B are invertible. Show that $(\delta A, \delta B)$ is a degenerate direction if and only if $\delta B = -B\,\delta A\,A^{-1}$. Count the number of dimensions in the subspace of degenerate directions.

(d) Now consider the general case. Fix A and B, and let $r_A = \operatorname{rank}(A)$ and $r_B = \operatorname{rank}(B)$. Show that the dimensionality of the space of degenerate directions is

$$m^2 + (m - r_A)(m - r_B).$$

Hint: The following is a guide to one possible approach.

(i) Describe the image of the linear map T_A sending $\delta B \in \mathbb{R}^{m^2}$ to $\delta B\,A \in \mathbb{R}^{m^2}$ in terms of the row space of A. Show that the dimensionality of this image (the rank of the linear map T_A) is mr_A.

(ii) Similarly, describe the image of the linear map T_B sending $\delta A \in \mathbb{R}^{m^2}$ to $B\,\delta A \in \mathbb{R}^{m^2}$ in terms of the column space of B. Show that the dimensionality of this image (the rank of the linear map T_B) is mr_B.

(iii) Describe the intersection of the images of T_A, T_B in terms of the row/column spaces of A, B. Show that the dimensionality of this intersection is $r_A r_B$.

(iv) Consider a third linear map, T_D, sending $(\delta A, \delta B) \in \mathbb{R}^{2m^2}$ to $B\,\delta A + \delta B\,A \in \mathbb{R}^{m^2}$. Describe the image of this linear map in terms of those of T_A and T_B. Compute the rank of this linear map using Grassmann's identity.

(v) What is the nullity of T_D? Why is this the same as the dimensionality of the space of degenerate directions?

Alternatively: substitute singular value decompositions $A = U_1\Sigma_1 V_1^\top$ and $B = U_2\Sigma_2 V_2^\top$ into the degeneracy condition $B\,\delta A + \delta B\,A = 0$, and change variables to absorb the orthogonal factors. The condition becomes an equation between two diagonally-scaled matrices, whose solutions can be counted entry by entry.

Exercise 2.10 (Redundant units in an MLP). Consider a single-hidden-layer MLP with scalar inputs and outputs, h hidden units with activation function $\sigma : \mathbb{R} \to \mathbb{R}$, and an output bias. The parameter space is $\mathcal{W} = \mathbb{R}^{3h+1}$ with

parameters $w = (a_1, b_1, c_1, \ldots, a_h, b_h, c_h, d)$, and the parameter–function map is

$$f_w(x) = d + \sum_{i=1}^{h} a_i \, \sigma(b_i x + c_i).$$

Here, for each hidden unit i, a_i is the outgoing weight, b_i is the incoming weight, and c_i is the bias. The output bias is d.

(a) Suppose $a_i = 0$ (unit i has zero outgoing weight). Show that the directions δb_i and δc_i are degenerate at w.

(b) Suppose $b_i = 0$ (unit i has zero incoming weight). Show that the direction $(\delta a_i, \delta d) = (1, -\sigma(c_i))$ (with all other components zero) is degenerate at w.

(c) Suppose $h \geq 2$ and $(b_i, c_i) = (b_j, c_j)$ for some $i \neq j$ (units i and j have the same incoming weights and biases). Show that the direction $(\delta a_i, \delta a_j) = (1, -1)$ (with all other components zero) is degenerate at w.

Remark (Localised symmetries in neural networks)

Each degenerate direction identified in the exercises above corresponds to a localised symmetry: a curve of functionally equivalent parameters that continuously extends only within a subset of parameter space.

1. In the DLN, the global change-of-basis symmetry $(A, B) \mapsto (GA, BG^{-1})$ for invertible $G \in \mathbb{R}^{m \times m}$ accounts for m^2 degenerate directions. These directions are present even when A and B have full rank. However, as we showed in Exercise 2.9, when A or B drops rank, $(m - r_A)(m - r_B)$ additional degenerate directions open up. These extra directions are localised to the rank-deficient region of parameter space.

2. In the MLP, at a parameter with $a_i = 0$, the path $t \mapsto (\ldots, a_i, b_i + t, c_i, \ldots)$ traces equivalent parameters within $\{a_i = 0\}$ (since f_w does not depend on b_i when $a_i = 0$), but this does not extend to a symmetry at nearby parameters with $a_i \neq 0$. Similarly, at a parameter with $(b_i, c_i) = (b_j, c_j)$, the path transferring weight between units i and j is a localised version of the sum symmetry from Exercise 2.5.

Remark (Rank of a neural network parameter)

In both exercises above, the degree of degeneracy at a parameter is controlled by the amount of redundant capacity in the network. For the two-layer DLN, this is measured by the rank of the product BA: a rank-r linear map can be implemented with a hidden dimension of r, leaving $m - r$ dimensions redundant. For a general single-hidden-layer MLP, this idea generalises in that we can define

the rank of a parameter w as the minimum number of hidden units needed to implement f_w (Farrugia-Roberts, 2022, 2024). In the linear case, this "neural network rank" coincides with the matrix rank of BA. In both settings, lower rank corresponds to a higher number of degenerate directions.

Remark (Measure zero degenerate sets)

When a parameter–function map is degenerate somewhere but not everywhere, it is often the case that it is degenerate only within a measure-zero subset of parameter space. This means that sampling parameters uniformly at random results in a degenerate parameter with probability zero. However, this does not mean that the degeneracies can be dismissed. The learning process applies a non-random selection pressure and may select parameters from a measure-zero set. Moreover, the existence of degenerate parameters can have practical consequences for nearby non-degenerate parameters, and the collective neighbourhoods of all degenerate parameters comprise a non-measure-zero subset of parameter space.

2.4 Degeneracy and information singularities

In the preceding sections, we studied degeneracy as a property of parameter–function maps. In the statistical framework introduced in Section 1.3, the fundamental object is instead a parameter–distribution map $\Psi : \mathcal{W} \to \mathcal{D}$ ($w \mapsto p_w : \mathcal{X} \to \Delta(\mathcal{Y})$). In this section, we extend the definition of degeneracy to parameter–distribution maps and connect it to a classical quantity from statistics, the Fisher information matrix. We will show that under mild regularity conditions on the statistical model, the Fisher information matrix is singular at w (has zero eigenvalues) if and only if the parameter–distribution map is degenerate at w.

Given a parameter–distribution map $\Psi : \mathcal{W} \to \mathcal{D}$, say that Ψ is **degenerate at w in direction v** if the conditional density does not change to first order:

$$D_v\Psi(w) = 0. \tag{21}$$

Here, $D_v\Psi(w)$ is not a conditional distribution but a signed difference between conditional distributions, effectively a real function of x and y (in particular, it may include negative density changes). We mean that this function is zero for all $x \in \mathcal{X}$ and all $y \in \mathcal{Y}$.

Equation (21) extends the definition of degeneracy from Equation (19). If Ψ is constructed from a parameter–function map Φ via a noise model (as in Section 1.3), then degeneracy of Φ at w in direction v implies degeneracy of Ψ at w in direction v, since $p(y \mid x, w)$ depends on w only through $f_w(x)$.

To relate degeneracy to the Fisher information matrix, we introduce two standard definitions.

Definition 2.3 (Score function)**.** Let $\Psi : \mathcal{W} \to \mathcal{D}$ be a parameter–distribution map with densities $p(y \mid x, w)$ that are positive and differentiable in w. The **score function** at w is

$$s(x, y, w) = \nabla_w \log p(y \mid x, w) \in \mathbb{R}^d. \tag{22}$$

The **directional score** in non-zero direction $v \in \mathbb{R}^d$ is

$$s_v(x, y, w) = D_v \log p(y \mid x, w) = \frac{v}{\|v\|} \cdot \nabla_w \log p(y \mid x, w). \tag{23}$$

The directional score measures how sensitive the log-density is to perturbations of w in direction v.

Exercise 2.11 (Basic properties of the score function). Let $\Psi : \mathcal{W} \to \mathcal{D}$ be a parameter–distribution map with positive densities $p(y \mid x, w)$, differentiable in w. Assume that for each x and w, the operations ∇_w and $\int_{\mathcal{Y}} \cdot \, dy$ may be exchanged.

(a) Show that the score function satisfies the following identity: for each $x \in \mathcal{X}$, $y \in \mathcal{Y}$, and $w \in \mathcal{W}$,

$$s(x, y, w) = \frac{\nabla_w p(y \mid x, w)}{p(y \mid x, w)}. \tag{24}$$

(b) Show that the expected score is zero: for each $x \in \mathcal{X}$ and $w \in \mathcal{W}$,

$$\mathbb{E}_{y \sim p(y|x,w)}\big[s(x, y, w)\big] = 0. \tag{25}$$

Hint: Differentiate the identity $\int_{\mathcal{Y}} p(y \mid x, w)\, dy = 1$.

(c) Show that if Ψ is degenerate at w in direction v, then the directional score vanishes identically: $s_v(x, y, w) = 0$ for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$.

(d) Show the converse: if the directional score $s_v(x, y, w)$ vanishes for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$ at w, then Ψ is degenerate at w in direction v.

Definition 2.4 (Fisher information matrix)**.** Let $\Psi : \mathcal{W} \to \mathcal{D}$ be a parameter–distribution map with score function s. Let $q(x)$ denote the marginal distribution of inputs. The **Fisher information matrix** $I : \mathcal{W} \to \mathbb{R}^{d \times d}$ is defined by

$$I(w) = \mathbb{E}_{x \sim q(x)}\, \mathbb{E}_{y \sim p(y|x,w)}\big[s(x, y, w)\, s(x, y, w)^\top\big]. \tag{26}$$

The Fisher information matrix is symmetric and positive semidefinite at every w (being an expectation of positive semidefinite matrices $s\, s^\top$). However, the Fisher information matrix may not be symmetric positive *definite,* that is, it may be singular.

The following exercise shows that singularity of the Fisher information matrix is equivalent to degeneracy of the parameter–distribution map under mild regularity conditions.

Exercise 2.12 (Fisher information and degeneracy). Let $\Psi : \mathcal{W} \to \mathcal{D}$ be a parameter–distribution map with densities $p(y \mid x, w)$. Assume that $p(y \mid x, w) > 0$ for all $y \in \mathcal{Y}$, $x \in \mathcal{X}$, $w \in \mathcal{W}$, that $w \mapsto p(y \mid x, w)$ is differentiable, and that $q(x) > 0$ for all $x \in \mathcal{X}$.

(a) Show that for any unit vector $v \in \mathbb{R}^d$,

$$v^\top I(w)\, v = \mathbb{E}_{x \sim q(x)}\, \mathbb{E}_{y \sim p(y|x,w)}\big[s_v(x, y, w)^2\big]. \tag{27}$$

(b) Using Exercise 2.11(c), show that if $D_v \Psi(w) = 0$ for some non-zero v, then $I(w)$ is not positive definite.

Hint: Check the definition of positive definite.

(c) Using Exercise 2.11(d), show that if $I(w)$ is not positive definite, then there exists a non-zero v for which $D_v \Psi(w) = 0$.

Hint: Check the definition of positive definite.

(d) Conclude that

$$\ker I(w) = \big\{v \in \mathbb{R}^d : D_v \Psi(w) = 0\big\}. \tag{28}$$

That is, the null space of the Fisher information matrix is exactly the space of degenerate directions of the parameter–distribution map.

Remark (Watanabe's strictly singular models)

Watanabe (2009) defines a statistical model as *strictly singular* if either of two conditions hold:

1. The Fisher information matrix $I(w)$ is singular for some $w \in \mathcal{W}$.
2. The parameter–distribution map is not one-to-one, that is, for some $w, w' \in \mathcal{W}$ such that $w \neq w'$, we have $\Psi(w) = \Psi(w')$.

By Exercise 2.12, under mild regularity conditions, the first condition is equivalent to our notion of (somewhere) degeneracy. The second condition is called non-identifiability.
Watanabe (2007, 2009) observe that many of the results from classical statistics crucially assume among their regularity conditions the parameter–distribution map is identifiable and the Fisher information matrix is nonsingular. Whereas, in a statistical model based on a non-trivial neural network (or any other statistical

model involving hierarchical structure), it is typical for the Fisher information matrix to include singularities.

2.5 Degeneracy and the loss landscape

In the preceding sections, we studied degeneracy as a property of parameter–function maps and parameter–distribution maps. We now consider the implications of degeneracy in these maps for the *loss landscape* in which deep learning algorithms operate.

Recall the definitions of loss functions from Section 1. In what follows, we assume that the per-example loss $L_{x,y}(w)$ depends on w only through $\Phi(w)$, and work with the population loss $L(w) = \mathbb{E}_{(x,y)\sim q}[L_{x,y}(w)]$. For statistical models, we use the expected negative log-likelihood $L(w) = \mathbb{E}_{(x,y)\sim q}[-\log p(y \mid x, w)]$ as our loss function. In either case, we assume that L is twice-differentiable with respect to w.

Let us begin with some basic observations about the relationship between degeneracy in parameter–function maps and directional derivatives in the loss landscape.

Exercise 2.13 (Directional derivatives of the loss). Let $L : \mathcal{W} \to \mathbb{R}$ be a population loss function satisfying the assumptions described above.

(a) Show that if Φ is degenerate at w in direction v, then the directional derivative of the loss vanishes in the same direction: $D_v L(w) = 0$.

(b) Suppose $\mathcal{W} = \mathbb{R}^d$. Show that if $d > 1$, then for *any* $w \in \mathcal{W}$, there exists a nonzero v such that $D_v L(w) = 0$.

Hint: Consider separately the cases $\nabla L(w) = 0$ and $\nabla L(w) \neq 0$.

We see that parameter–function map degeneracy implies flat loss directions, but a vanishing directional derivative of the loss landscape is commonplace. To find a satisfying definition of degeneracy in a loss landscape, we must look at second-order information.

Define the **Hessian matrix** $H(w) \in \mathbb{R}^{d\times d}$ of the loss at w by

$$H(w)_{jk} = \frac{\partial^2 L}{\partial w_j \partial w_k}(w). \tag{29}$$

The Hessian at w is sometimes denoted $\nabla^2 L(w)$. The Hessian captures the local curvature of the loss landscape via the quadratic approximation

$$L(w + \delta) \approx L(w) + \nabla L(w)^\top \delta + \tfrac{1}{2}\, \delta^\top H(w)\, \delta. \tag{30}$$

At a critical point ($\nabla L(w) = 0$), the Hessian determines whether the loss curves upward, downward, or remains flat in each direction.

In particular, at a local minimum, $H(w)$ is positive semidefinite. We can therefore classify local minima as follows:

- A local minimum w is **regular** (or **non-degenerate**, or **Morse**) if $H(w)$ is positive definite.
- A local minimum w is **degenerate** (or **non-Morse**) if $H(w)$ is singular (has a zero eigenvalue).

Exercise 2.14 (Some examples of loss landscape degeneracy). Consider the parameter space $\mathcal{W} = \mathbb{R}^2$ with the identity parameter–function map $\Phi = \mathrm{id}$, so that we identify parameters with the functions they implement (cf., Exercise 2.1). Consider the family of loss functions $L_{k,l}(a, b) = a^{2k} + b^{2l}$ for non-negative integers k and l.

(a) Show that the origin is a global minimum of $L_{k,l}$ for all non-negative k and l. For which k and l is it the unique global minimum?

(b) Show that if $k = l = 1$, then the origin is a non-degenerate (regular) minimum.

Hint: Compute the Hessian.

(c) Show that if $k > 1$ and $l \geq 1$, then the origin is a degenerate minimum.

(d) Show that if $k = 0$ and $l \geq 1$, then the origin is a degenerate minimum.

Note that the Hessian degeneracy in Exercises 2.14(c) and 2.14(d) arises for qualitatively different reasons. In Exercise 2.14(d), the loss is constant along the a-direction to all orders—a genuinely flat direction. In Exercise 2.14(c), the loss does increase along the a-direction, just more slowly than quadratically (as a^4, a^6, etc., depending on k). Moreover, different values of k give different rates of increase. The Hessian cannot distinguish any of these cases: flat, quartic, and so on all appear the same from the perspective of the Hessian rank. We will later see approaches that can distinguish these degrees of degeneracy.

So much for defining loss landscape degeneracy. What is the relationship between this kind of degeneracy and degeneracy of the parameter–function/distribution map?

The relationship is subtle, and generally depends on the choice of loss function. One natural setting in which to study this relationship is the *realisable statistical model:* given a parameter–distribution map and a parameter w_0, if we consider p_{w_0} as the true data-generating distribution, the population negative log-likelihood loss will have a global minimum at w_0. The following exercise shows that in this setting, under mild regularity assumptions on the statistical model, degeneracy in the parameter–distribution map at w_0 implies that the global minimum w_0 is a degenerate global minimum of the loss landscape.

Exercise 2.15 (Realisable models and Hessian degeneracy). Let $\Psi : \mathcal{W} \to \mathcal{D}$ be a parameter–distribution map with positive densities $p(y \mid x, w)$, twice-differentiable in w. Suppose data is generated from a fixed true parameter $w_0 \in \mathcal{W}$, meaning $q(y \mid x) = p(y \mid x, w_0)$. Consider the population negative log-likelihood loss

$$L(w) = -\mathbb{E}_{x\sim q(x)}\, \mathbb{E}_{y\sim p(y|x,w_0)}\big[\log p(y \mid x, w)\big]. \tag{31}$$

Assume that for each x, the operations ∇_w (and ∇_w^2) and $\int_{\mathcal{Y}} \cdot\, dy$ may be exchanged.

(a) *(Bartlett identity.)* Show that for each $x \in \mathcal{X}$ and $w \in \mathcal{W}$,

$$-\mathbb{E}_{y\sim p(y|x,w)}\big[\nabla_w^2 \log p(y \mid x, w)\big] = \mathbb{E}_{y\sim p(y|x,w)}\big[s(x, y, w)\, s(x, y, w)^\top\big], \tag{32}$$

where s is the score function from Definition 2.3.

Hint: Differentiate the identity $\mathbb{E}_{y\sim p(y|x,w)}[s(x, y, w)] = 0$ (established in Exercise 2.11) with respect to w.

(b) Show that the Hessian of L at the true parameter w_0 equals the Fisher information matrix (Definition 2.4):

$$H(w_0) = I(w_0). \tag{33}$$

Hint: Compute $H(w) = -\mathbb{E}_x\, \mathbb{E}_{y\sim p(y|x,w_0)}[\nabla_w^2 \log p(y \mid x, w)]$, evaluate at $w = w_0$, and apply part (a).

(c) Using the result of Exercise 2.12, conclude: if Ψ is degenerate at w_0 in direction v, then $H(w_0)\, v = 0$. Contrapositively, if $H(w_0)$ is positive definite, then Ψ is non-degenerate at w_0.

Exercise 2.15 shows that for the population negative log-likelihood at the true parameter, degeneracy of the parameter–distribution map implies a degenerate loss landscape. The converse does not hold in general, nor does the forward direction hold at arbitrary points in parameter space. The following exercise explores these subtleties through examples.

Exercise 2.16 (Contrasting parametric versus loss degeneracy).

(a) Observe that the identity parameter–function map on $\mathcal{W} = \mathbb{R}^2$ is non-degenerate everywhere. With this in mind, what do your answers to Exercises 2.14(b) to 2.14(d) exemplify about the logical relationship between parameter–function map degeneracy and loss landscape degeneracy?

(b) Consider again the identity parameter–function map on $\mathcal{W} = \mathbb{R}^2$. Consider the loss function $L_\circ(a, b) = (a^2 + b^2 - 1)^2$. Find a global minimum of $L_\circ$ and show that it is a degenerate critical point. What does this example reveal about the logical relationship between parameter–function map degeneracy and loss landscape degeneracy?

(c) Now consider the cubic parameter–function map $\Phi(\alpha, \beta) = (\alpha^3, \beta^3)$.

(i) Argue that Φ is degenerate at the origin (cf., Exercise 2.2).

(ii) If we take $L_{k,l}$ from Exercise 2.14 to be defined on the constant function (a, b) (rather than the equivalent underlying parameter), then in this parameterisation, that loss function becomes $L_{k,l}(\alpha, \beta) = (\alpha^3)^{2k} + (\beta^3)^{2l}$. Show that the origin is a degenerate minimum of $L_{k,l}$ for all $k \geq 0$ and $l \geq 0$ under this parametrisation.

(iii) What does this example reveal about the logical relationship between parameter–function map degeneracy and loss landscape degeneracy?

(d) Consider the product parameter–function map $\Phi(a, b) = a \cdot b$ from Exercise 2.3, and define $L(a, b) = \frac{1}{2}(ab - 1)^2$.

(i) Show that $(0, 0)$ is a critical point and that Φ is degenerate in every direction at $(0, 0)$.

(ii) Compute the Hessian at $(0, 0)$ and show it is nonsingular.

(iii) What does this example reveal about the logical relationship between parameter–function map degeneracy and loss landscape degeneracy? Why does this not contradict Exercise 2.15?

3 The degeneracy hierarchy

We have so far explored different kinds of qualitative degeneracy in the loss landscape. A natural question arises: *how can we quantify the degree of degeneracy at a particular point in parameter space?* In this section, we introduce the local learning coefficient, a rich mathematical object that reflects the degree of degeneracy at a parameter, and we explore it from several perspectives.

In this section and Section 4 we assume that any population loss L is a real analytic function. The precise definition of real analyticity is not important for this tutorial but interested readers can refer to Watanabe (2009) (section 2). A property that will be useful is that such functions have a convergent Taylor expansion, in particular they have smooth derivatives of all orders.

3.1 The local learning coefficient via volume scaling asymptotics

The key idea is to measure the *volume* of near-optimal parameters. Consider a local minimum $w^* \in \mathcal{W}$ of the population loss L, and suppose we have a sufficiently small closed ball $B(w^*)$ centred on w^* such that $L(w) \geq L(w^*)$ for all $w \in B(w^*)$. Given a tolerance $\epsilon > 0$, define the **sublevel set**

$$B(w^*, \epsilon) = \{w \in B(w^*) \mid L(w) - L(w^*) < \epsilon\}, \tag{34}$$

consisting of all parameters in the ball whose loss is within ϵ of the minimum. The volume of this set is

$$V(\epsilon) := \mathrm{Vol}(B(w^*, \epsilon)) = \int_{B(w^*,\epsilon)} dw. \tag{35}$$

We will look at how this volume scales as $\epsilon \to 0$. It will be instructive for us to first calculate this scaling in the non-degenerate case.

Exercise 3.1 (Volume scaling in the regular case). Let $w^* \in \mathcal{W}$ be a local minimum of L, and suppose that the Hessian $\nabla^2 L(w^*)$ is positive definite. Show that

$$V(\epsilon) = c\,\epsilon^{d/2} + o(\epsilon^{d/2}) \quad \text{as } \epsilon \to 0, \tag{36}$$

for some constant $c > 0$ where $d = \dim(\mathcal{W})$. You may assume that, as $\epsilon \to 0$, the volume of $B(w^*, \epsilon)$ agrees to leading order with the volume of the quadratic approximation obtained by replacing L with its second-order Taylor expansion at w^*.

Hint: The volume of the ellipsoid $\{\boldsymbol{x} \in \mathbb{R}^d \mid \boldsymbol{x}^\top A \boldsymbol{x} < 1\}$ is proportional to $(\det A)^{-1/2}$.

The key takeaway from this exercise is that the dimension of the model controls the leading order asymptotics of the volume scaling. We can investigate this further by calculating the volume scaling in an example where the Hessian is not positive definite.

Exercise 3.2 (Quadratic Valley Volume Scaling). Let $\mathcal{W} = [-R, R]^d$ for some (large) $R \in \mathbb{R}$ and let $L(x_1, \ldots, x_d) = \sum_{i=1}^{d'} x_i^2$.

(a) Sketch the loss landscape when $d = 2$, $d' = 1$.

(b) L achieves its minimum value 0 at multiple points in $\mathcal{W}$. What is the dimension of the space $\mathcal{W}_0 := \{w \in \mathcal{W} \mid L(w) = 0\}$?

(c) Calculate the Hessian of L.

(d) Prove that $V(\epsilon) \propto \epsilon^{d'/2}$

Hint: Use the volume of the ellipsoid from Exercise 3.1.

In this example, the degenerate directions do not contribute to the leading exponent of $V(\epsilon)$. We see that the exponent $\frac{d}{2}$ from Exercise 3.1 is replaced with $\frac{d'}{2}$ where d' is the codimension of $\mathcal{W}_0$ i.e. the *effective dimension* of L which ignores the degenerate directions. This suggests that we can capture information about the degree of degeneracy of a loss function by looking at how the volume $V(\epsilon)$ scales as $\epsilon \to 0$. Watanabe (2009) proves a general form for this kind of volume scaling which has been adapted by Lau et al. (2025) into the following definition.

Definition 3.1 (Local learning coefficient)**.** Let $w^* \in \mathcal{W}$ be a local minimum of the population loss L. There exists a unique rational number $\lambda(w^*) > 0$, a positive integer $\mu(w^*)$, and a constant $c > 0$ such that as $\epsilon \to 0$,

$$V(\epsilon) = c\,\epsilon^{\lambda(w^*)}(-\log \epsilon)^{\mu(w^*)-1} + o\left(\epsilon^{\lambda(w^*)}(-\log \epsilon)^{\mu(w^*)-1}\right). \tag{37}$$

We call $\lambda(w^*)$ the **local learning coefficient (LLC)** at w^*, and $\mu(w^*)$ the **local multiplicity**.

Exercise 3.3 (Two equations for λ).

(a) Prove that for $a > 1$

$$\lambda = \lim_{\epsilon \to 0} \frac{\log(V(a\epsilon)/V(\epsilon))}{\log(a)} \tag{38}$$

(b) Prove that

$$\lambda = \lim_{\epsilon \to 0} \frac{\log V(\epsilon)}{\log \epsilon} \tag{39}$$

Note that from equation (36) we already have that the LLC and local multiplicity in

the regular case are $\lambda = \frac{d}{2}$, $\mu = 1$.

When $\mu(w^*) = 1$, the formula simplifies to $V(\epsilon) = c\epsilon^{\lambda(w^*)} + o(\epsilon^{\lambda(w^*)})$ which allows us to interpret $\lambda(w^*)$ as the *volume scaling exponent*: increasing the error tolerance by a factor of a increases the volume of near-optimal parameters by a factor of $a^{\lambda(w^*)}$.

Exercise 3.4 (Calculating the LLC). Calculate the LLC and the rank of the Hessian at the global minimum in the following cases:

(a) $L(x) = x^2$

(b) $L(x) = x^{2m}$

(c) $L(x, y) = x^4 + y^4$

(d) $L(x, y) = x^2 + y^4$

Exercise 3.5 (Upper bound on the LLC). In Exercise 3.1, we saw that $\lambda = d/2$ for regular models. In this exercise, you will prove directly from the volume scaling definition that for any local minimum w^* of the population loss, the local learning coefficient satisfies $\lambda(w^*) \leq d/2$.

(a) Let $\Lambda > 0$ be an upper bound on the eigenvalues of the Hessian $\nabla^2 L(w)$ for all $w \in B(w^*)$. Using the Lagrange remainder form of Taylor's theorem, show that for all $w \in B(w^*)$,

$$L(w) - L(w^*) \leq \frac{1}{2}\Lambda \|w - w^*\|^2.$$

Hint: For any real symmetric matrix $H \in \mathbb{R}^{n \times n}$ and vector $v \in \mathbb{R}^n$, $v^\top H v \leq \Lambda \|v\|^2$ where Λ is the largest eigenvalue of H.

(b) Consider the standard Euclidean ball of radius r centred at w^*, denoted $B_r(w^*) = \{w \in \mathcal{W} \mid \|w - w^*\| < r\}$. Find a radius r (as a function of ϵ and Λ) such that for sufficiently small ϵ,

$$B_r(w^*) \subseteq B(w^*, \epsilon).$$

(c) Recall that the volume of a d-dimensional Euclidean ball of radius r is proportional to r^d. Use your result from part (b) to show that there exists a constant $C > 0$ such that for sufficiently small ϵ,

$$C\epsilon^{d/2} \leq V(\epsilon).$$

(d) Conclude that $\lambda(w^*) \leq d/2$.

We have now shown that the LLC satisfies $\lambda(w^*) \leq d/2$, with equality precisely in the regular (non-degenerate) case. It is also possible (although quite fiddly) to prove that $\frac{r}{2} \leq \lambda(w^*)$ where $r = \text{rank}\nabla^2 L(w^*)$ and as we will see in the next exercise, the LLC captures more information about the geometry than just the Hessian rank.

Exercise 3.6 (LLC vs Hessian Rank). In this exercise we will show that the LLC can detect differences in the local geometry that are not reflected in the rank of the Hessian.
Construct two loss functions L_1 and L_2 with minima w_1 and w_2 respectively such that $\text{rank}\,\nabla^2 L_1(w_1) = \text{rank}\,\nabla^2 L_2(w_2)$ but $\lambda(w_1) \neq \lambda(w_2)$.

Exercise 3.7 (The cubically-parameterised loss). We can make a quadratic loss degenerate by changing the parameterisation. Let the *cubically-parameterised loss* be

$$L(\mu) = \tfrac{1}{2}(\mu^3 - \mu_0^3)^2, \qquad \mu \in \mathbb{R} \tag{40}$$

obtained from the ordinary quadratic loss $\ell(\theta) = \frac{1}{2}(\theta - \theta_0)^2$ via the reparameterisation $\theta = \mu^3$, $\theta_0 = \mu_0^3$.

(a) Show that the cubically-parameterised loss is just as expressive as the ordinary quadratic loss: that is, they both have the same set of achievable minima.

(b) Compute the Hessian $L''(\mu_0)$ to show that the loss is degenerate at its minimum when $\mu_0 = 0$, and non-degenerate when $\mu_0 \neq 0$.

(c) Give an explicit formula for $V(\epsilon)$ when $\mu_0 = 0$ and $\mu_0 \neq 0$ separately.

(d) Use your answer to part (c) to find the learning coefficient for $\mu_0 = 0$ and for $\mu_0 \neq 0$. Given that the model is non-degenerate when $\mu_0 \neq 0$, we expect the learning coefficient to be $d/2$ in that case; compare your answer. How does the cubically-parameterised loss at $\mu_0 = 0$ differ from the ordinary quadratic loss?

(e) Instead of taking $\epsilon \to 0$ to get the learning coefficient, fix a small but nonzero value for ϵ, such as $\epsilon = 0.01$. Plot $V(\epsilon)$ as a function of μ_0. As we saw from (d), the learning coefficient changes discontinuously when $\mu_0 = 0$—what happens with $V(\epsilon)$ as μ_0 gets close to zero? What changes if you make ϵ smaller or larger?

Even though the asymptotic *learning coefficient* ($\epsilon \to 0$) only changes when $\mu_0 = 0$ exactly, note how the non-asymptotic *volume* (ϵ finite) is affected in a larger neighbourhood.

3.2 Perspectives on the local learning coefficient

In this section, we survey several alternative perspectives on the LLC, from information theory, algebraic geometry, and geometric measure theory.

Information-theoretic interpretation. The LLC also admits a natural interpretation in terms of information (Lau et al., 2025). The number of bits needed to specify the sublevel set $B(w^*, \epsilon)$ within the ball $B(w^*)$ is

$$-\log_2 \frac{V(\epsilon)}{\mathrm{Vol}(B(w^*))}. \tag{41}$$

For small ϵ and $\mu(w^*) = 1$, this is approximately $-\lambda(w^*)\log_2 \epsilon + O(\log_2 \log_2 \epsilon)$. In particular, the number of *additional* bits needed to halve an already small error tolerance from ϵ to $\epsilon/2$ is

$$-\log_2 \frac{V(\epsilon/2)}{V(\epsilon)} \approx \lambda(w^*). \tag{42}$$

That is, the LLC $\lambda(w^*)$ measures the number of bits required to halve the error near w^*.

Local learning coefficient via algebraic geometry We explore an algebro-geometric perspective on the learning coefficient due to Watanabe (2009). Note that this section is more advanced than the remainder of this tutorial, and should be considered optional. When $L(w)$ is a real analytic function, Hironaka's resolution of singularities theorem guarantees the existence of a smooth map $g\colon M \to \mathcal{W}$ from an analytic manifold M and local coordinates $u = (u_1, \ldots, u_d)$ that *monomialise* the loss near w^*:

$$L(g(u)) - L(w^*) = u_1^{2k_1} \cdots u_d^{2k_d}, \qquad dw = b(u)\, u_1^{h_1} \cdots u_d^{h_d}\, du,$$

where $b(u) > 0$ is smooth and the exponents k_j, h_j are non-negative integers. In these coordinates, the volume integral $V(\epsilon)$ can be evaluated explicitly, yielding the asymptotic form of Definition 3.1 and giving

$$\lambda(w^*) \;=\; \min_{\text{coordinate charts}} \; \min_{j\colon k_j > 0} \frac{h_j + 1}{2k_j}\,. \tag{43}$$

The resolution map g is far from unique, and different choices produce different exponents k_j, h_j. $\lambda(w^*)$ is nevertheless well-defined, which can be seen from (another) equivalent definition using the **local zeta function**:

$$\zeta(z,\, w^*) \;=\; \int_{B(w^*)} \big(L(w) - L(w^*)\big)^z \, dw\,, \tag{44}$$

which converges for $\mathrm{Re}(z) > 0$ and extends to a meromorphic function on $\mathbb{C}$ whose poles are all real, negative, and rational. The LLC $\lambda(w^*)$ is precisely the negative of

the largest pole, and the multiplicity $\mu(w^*)$ is the multiplicity of this pole. Since (44) makes no reference to any resolution, $\lambda(w^*)$ depends only on L and w^*.

In the algebraic geometry literature, $\lambda(w^*)$ is called the **real log canonical threshold (RLCT)** of L at w^*. We collect several consequences of this perspective.

1. The fact that the LLC is a rational number follows directly from equation (43).
2. Equation (43) suggests that we can compute $\lambda(w^*)$ analytically by constructing an explicit resolution of singularities for $L(w) - L(w^*)$ and reading off the exponents. This is difficult in practice for neural network loss functions, but has been achieved for some architectures, as we will see in Section 3.3.
3. In Section 4 we will state Watanabe's free energy formula connecting the Bayesian free energy to the LLC. The proof of this result makes repeated use of the monomialisation of the loss.

Exercise 3.8 (Computing the LLC via the RLCT). Suppose that $L(w) = w_1^{2k_1} \cdots w_d^{2k_d}$ near $w^* = 0$, where $k_j \geq 0$ are integers with at least one $k_j > 0$. Show that

$$\lambda(0) = \min_{j:\, k_j > 0} \frac{1}{2k_j}.$$

Local learning coefficient as a fractal dimension The LLC admits a geometric interpretation: it is a *fractal dimension* of the parameter space, as measured through the lens of the loss function. To make this precise, we recall a standard notion from fractal analysis.

Definition 3.2 (Hölder exponent)**.** Let (X, ρ) be a (pseudo-)metric space and ν a measure on X. The **Hölder exponent** (or **local dimension**) $\alpha(x)$ of ν at a point $x \in X$ is given by

$$\alpha(x) = \lim_{\epsilon \to 0} \frac{\log \nu(B_\epsilon(x))}{\log \epsilon}, \tag{45}$$

where $B_\epsilon(x)$ is the ball of radius ϵ centred at x. Equivalently, $\alpha(x)$ is the unique real number such that

$$\nu(B_\epsilon(x)) = O(\epsilon^{\alpha(x)}) \qquad \text{as } \epsilon \to 0.$$

The Hölder exponent generalises the notion of dimension: when ν is the Lebesgue measure on $\mathbb{R}^d$ and ρ is the Euclidean metric, every point has $\alpha(x) = d$, but for measures concentrated on fractal sets the exponent can be non-integer. Notice that the formula (45) has a similar form to the definition of the local learning coefficient (38). To make this connection precise, we introduce a pseudo-metric induced by the loss function.

Definition 3.3 (Loss pseudo-metric)**.** The **loss pseudo-metric** on $\mathcal{W}$ is defined by

$$\rho(w, u) = |L(w) - L(u)|. \tag{46}$$

Remark

ρ is only a pseudo-metric as the distance between two distinct points can be zero.

Under this pseudo-metric, the sublevel set (34) is precisely the ϵ-ball around w^*:

$$B(w^*, \epsilon) = \{w \in B(w^*) \mid \rho(w, w^*) < \epsilon\}, \tag{47}$$

so $V(\epsilon) = \nu(B(w^*, \epsilon))$.

Exercise 3.9 (Hölder exponent and learning coefficient). Let $\alpha(w^*)$ be the Hölder exponent of the Lebesgue measure on $\mathcal{W}$ at w^* under the loss pseudo-metric. Show that

$$\lambda(w^*) = \alpha(w^*).$$

3.3 Local learning coefficients of deep linear networks

Having established the local learning coefficient as a measure of degeneracy in the loss landscape, we now compute it explicitly for the two-layer deep linear network introduced in Section 1. In Section 2, we saw that DLN parameters exhibit varying degrees of degeneracy depending on the ranks of the component matrices (Exercise 2.9, Remark 2.2). The LLC makes this hierarchy quantitative: different ranks correspond to different learning coefficients, confirming that lower-rank parameters are geometrically "simpler."

Consider the two-layer DLN $f_{A,B}(x) = BAx$ with $A \in \mathbb{R}^{H \times M}$ and $B \in \mathbb{R}^{N \times H}$, so the parameter space is $\mathcal{W} = \mathbb{R}^{(M+N)H}$. Suppose the true input–output relationship is $y = B_0 A_0 x$ where $A_0 \in \mathbb{R}^{H \times M}$, $B_0 \in \mathbb{R}^{N \times H}$, and $r = \text{rank}(B_0 A_0)$, and consider the population loss

$$L(A, B) = \mathbb{E}_{x \sim q(x)} \left[\|BAx - B_0 A_0 x\|^2 \right], \tag{48}$$

where $q(x)$ has full-rank covariance. The set of minima of this loss function is $W_0 = \{(A, B) : BA = B_0 A_0\}$. Aoyagi and Watanabe (2005) derived an exact formula for the learning coefficient at these minima.

Theorem 3.4 (Aoyagi and Watanabe, 2005)**.** *For the two-layer DLN $f_{A,B}(x) = BAx$ with $A \in \mathbb{R}^{H \times M}$, $B \in \mathbb{R}^{N \times H}$, and true parameter (A_0, B_0) with $r = \text{rank}(B_0 A_0)$, at any minimum $w^* \in W_0$ the local learning coefficient λ and its multiplicity μ are given by the following cases:*

1. *If $N + r \leq M + H$, $M + r \leq N + H$, and $H + r \leq M + N$:*

(a) *If $M + H + N + r$ is even, then $\mu = 1$ and*

$$\lambda = \frac{-(H+r)^2 - M^2 - N^2 + 2(H+r)M + 2(H+r)N + 2MN}{8}.$$

(b) *If $M + H + N + r$ is odd, then $\mu = 2$ and*

$$\lambda = \frac{-(H+r)^2 - M^2 - N^2 + 2(H+r)M + 2(H+r)N + 2MN + 1}{8}.$$

2. *If $M + H < N + r$, then $\mu = 1$ and $\lambda = \dfrac{HM - Hr + Nr}{2}$.*

3. *If $N + H < M + r$, then $\mu = 1$ and $\lambda = \dfrac{HN - Hr + Mr}{2}$.*

4. *If $M + N < H + r$, then $\mu = 1$ and $\lambda = \dfrac{MN}{2}$.*

Exercise 3.10 (Learning coefficients of two-layer DLNs). Consider the constant-width two-layer DLN from Exercise 2.9: $y = BAx$, where $x \in \mathbb{R}^m$ are the inputs, $y \in \mathbb{R}^m$ are the outputs, and $A, B \in \mathbb{R}^{m \times m}$ are linear transformations. The parameter space is $\mathcal{W} = \mathbb{R}^{2m^2}$ and the parameters are the pair $(A, B) \in \mathcal{W}$.

(a) Aoyagi & Watanabe do not assume the network is constant-width, like we do. Simplify their formula for λ using this assumption.

(b) How does the learning coefficient λ depend on r, the rank of $B_0 A_0$? Compare with the results of Exercise 2.9(d).

(c) How does λ compare with the effective parameter count $d/2$ of a regular model? What does this tell us about the role of singularities in the two-layer DLN?

4 Degeneracy and Bayesian deep learning

In Sections 2 and 3, we studied the geometry of degeneracy in parameter space and introduced the local learning coefficient $\lambda(w^*)$ as a measure of the degree of degeneracy at a local minimum w^*. We now show that this geometry has consequences for learning (in the Bayesian setting). In particular, it creates a trade-off between model fit and model complexity that drives learning, providing a mechanism for *internal model selection*.

4.1 Watanabe's free energy formula

Recall the Bayesian learning setup from Section 1.4. We have a parameter–distribution map $\Psi : \mathcal{W} \to \mathcal{D}$ sending each parameter w to a conditional distribution with density $p(y \mid x, w)$, a smooth positive prior φ on $\mathcal{W}$, and a data set of n i.i.d. examples from a data distribution q. Take the loss function to be the negative log-likelihood

$$L_n(w) = -\frac{1}{n}\sum_{i=1}^{n} \log p(y^{(i)} \mid x^{(i)}, w), \qquad L(w) = \mathbb{E}_{(x,y)\sim q}[-\log p(y \mid x, w)]\,.$$

The posterior π_n is given by Bayes' rule (13) and as we saw in Exercises 1.7 and 1.8 it concentrates around regions of parameter space with the lowest local free energy $F_n(\mathcal{U})$ (Definition 1.6).

We now state one of the main results of SLT: an asymptotic expansion of $F_n(\mathcal{U})$ as $n \to \infty$. Under certain technical assumptions on the statistical model (Watanabe, 2018; see Lau, 2025, §2.3.1 for a summary), we have the following theorem.

Theorem 4.1 (Local free energy formula)**.** *Suppose $\mathcal{U} \subseteq \mathcal{W}$ contains at least one minimiser of L, and let $w^* \in \mathcal{U}$ be any such minimiser. Define $\lambda_{\mathcal{U}}$ as the smallest local learning coefficient among all minimisers in $\mathcal{U}$, and let $\mu_{\mathcal{U}}$ be the maximum local multiplicity associated with $\lambda_{\mathcal{U}}$. Then, as $n \to \infty$,*

$$F_n(\mathcal{U}) = nL_n(w^*) + \lambda_{\mathcal{U}} \log n - (\mu_{\mathcal{U}} - 1)\log\log n + O_p(1). \tag{49}$$

Remark

Since $L_n(w^*)$ is a random variable, so is $F_n(\mathcal{U})$. The expansion Equation (49) is an asymptotic statement about random variables where the remainder $O_p(1)$ denotes a term that is bounded in probability as $n \to \infty$.

To understand what this expansion tells us about Bayesian learning, note that the two leading terms of this expansion have transparent interpretations:

$$F_n \approx \underbrace{nL_n}_{\text{data fit}} + \underbrace{\lambda \log n}_{\text{complexity}}$$

Let $w_1, w_2 \in \mathcal{W}$ be minimisers of the population loss with $L(w_1) = L(w_2)$ and $\lambda(w_1) \leq \lambda(w_2)$. Then for fixed large n, the free energy of a neighbourhood of w_1 is smaller than that of w_2, so the posterior concentrates around the simpler solution w_1.

This is an example of **internal model selection** where there are two equally good solutions that minimise the population loss. However, our learning algorithm (Bayesian updating) prefers one solution over the other, in particular the one with lower complexity.

Remark (Regular models and the BIC)

When the Hessian $\nabla^2 L(w^*)$ is non-degenerate, $\lambda(w^*) = d/2$ and $\mu(w^*) = 1$ by Exercise 3.1. The free energy formula then reduces to $F_n \approx nL_n(w^*) + \frac{d}{2}\log n$, recovering the Bayesian Information Criterion from classical statistics.

Exercise 4.1 (Numerical demonstration of the free energy formula). Consider the statistical model underlying the cubically-parameterised loss from Exercise 3.7. The parameter $\mu \in \mathbb{R}$ indexes the family of normal distributions

$$p(x \mid \mu) = (2\pi)^{-1/2} \exp\left(-\frac{1}{2}(x - \mu^3)^2\right).$$

Suppose the true distribution is

$$q(x) = (2\pi)^{-1/2} \exp\left(-\frac{1}{2}(x - \mu_0^3)^2\right)$$

for some fixed μ_0, and equip the model with a standard normal prior

$$\varphi(\mu) = (2\pi)^{-1/2} \exp\left(-\frac{1}{2}\mu^2\right).$$

[*Note:* The following exercises are intended to be attempted with a numerical programming framework such as Python or Julia, rather than by hand.]

(a) For $\mu_0 = 0$, numerically compute the free energy F_n for many values of n between $n = 100$ and $n = 10{,}000$. Compare the computed values against the asymptotic estimate $F_n \approx nL_n + \lambda \log n$, using the value of λ from Exercise 3.7(d).

(b) Repeat part (a) with $\mu_0 = 5$.

(c) Repeat part (a) with $\mu_0 = 0.1$. Note that $\mu_0 = 0.1 \neq 0$, so asymptotically $\lambda = 1/2$, as in part (b). However, observe how for moderate n the free energy behaves more like part (a) than part (b). This is another manifestation of the phenomenon from Exercise 3.7(e): the effects of the singularity at $\mu = 0$ extend into a neighbourhood around it.

4.2 Bayesian phase transitions

So far we have looked at the consequence of the free energy formula when considering n to be fixed. However, we also find it interesting to see what it tells us about changes in the posterior as n increases.

For the rest of this section, assume n is sufficiently large that (i) the free energy can be approximated by the first two terms in its expansion, and (ii) $L_n \approx L$ by the law of large numbers, so that the empirical loss can be treated as approximately constant in n.

Consider two disjoint regions $\mathcal{U}, \mathcal{V} \subseteq \mathcal{W}$, containing minimisers u^* and v^* of the population loss L with local learning coefficients $\lambda_{\mathcal{U}}$ and $\lambda_{\mathcal{V}}$ respectively. Applying the free energy formula to each region, the log posterior odds between the two regions satisfy

$$\log \frac{\pi_n(\mathcal{U})}{\pi_n(\mathcal{V})} = F_n(\mathcal{V}) - F_n(\mathcal{U}) \approx n\,\Delta L_n + \Delta\lambda \cdot \log n \tag{50}$$

where $\Delta L_n = L_n(v^*) - L_n(u^*)$ and $\Delta\lambda = \lambda_{\mathcal{V}} - \lambda_{\mathcal{U}}$.

Consider the case when one region is more accurate but more complex than the other, i.e.

$$L_n(u^*) > L_n(v^*) \qquad \text{and} \qquad \lambda_{\mathcal{U}} < \lambda_{\mathcal{V}}.$$

Then $\Delta L_n < 0$ and $\Delta\lambda > 0$, so the two terms in (50) have opposite signs. For smaller n, the $\Delta\lambda \cdot \log n$ term dominates so the log posterior odds is negative meaning the posterior prefers $\mathcal{U}$ (the simpler, less accurate region). As n increases, the linear term $n\,\Delta L_n$ eventually dominates the logarithmic term, and the posterior shifts to prefer $\mathcal{V}$ (the more complex, more accurate region). Setting the leading terms equal, this change occurs at a critical sample size n^* satisfying

$$\frac{n^*}{\log n^*} \approx \frac{\Delta\lambda}{|\Delta L_n|}. \tag{51}$$

At $n = n^*$, the posterior undergoes a **Bayesian phase transition**: the region that dominates the posterior changes abruptly.

Exercise 4.2 (Exploring phase transitions). We show how *phase transitions* can occur when different local free energies change at different rates.

(a) Suppose we partition the overall parameter space into a disjoint union $\mathcal{W} = \mathcal{U} \sqcup \mathcal{V}$. Denote the overall free energy by F_n and the local free energies by $F_n(\mathcal{U})$ and $F_n(\mathcal{V})$, respectively. Show that the relationship

$$F_n = -\log(e^{-F_n(\mathcal{U})} + e^{-F_n(\mathcal{V})})$$

holds.

(b) Suppose $F_n(\mathcal{U}) \approx 0.3n + 20\log n$ and $F_n(\mathcal{V}) \approx 0.5n + 2\log n$. Plot the overall

free energy $F(n)$. Compare against $\min\{F_n(U), F_n(\mathcal{V})\}$. What happens around $n = 570$?

(c) In statistical physics, a phase transition is traditionally defined as a discontinuity (or rapid change, for finite-size systems) in the derivatives of the free energy. Plot $\frac{d}{dn}F_n$ and explain why this justifies calling the phenomenon from b) a phase transition.

(d) Change the coefficients of the terms in b). How does this change when a phase transition occurs? Does a phase transition always occur?

5 Further readings

We conclude by providing several pointers to additional literature and resources for those interested in investigating singular learning theory (SLT) in more detail.

5.1 Other introductions to singular learning theory

For alternative introductions to SLT, see the following.

- Wei et al. (2023) "Deep Learning is Singular, and That's Good," a technical position paper surveying some implications of SLT for deep learning.
- Carroll (2023) "Distilling singular learning theory," a LessWrong sequence introducing Watanabe's free energy formula and discussing an example of a Bayesian phase transition in a small neural network.
- Lecture recordings from the *SLT & Alignment Summit, 2023*. In particular:
  - See the "SLT Low Road" lectures (Lau and Chen, 2023) for an outline of the derivation of Watanabe's free energy formula.
  - See the "SLT High Road" lectures (Murfet and Carroll, 2023) for a discussion of the free energy formula's implications including Bayesian phase transitions.
- Lau (2025, Chapter 2, "Singular Learning Theory Background"), a self-contained technical introduction to the main results of SLT with illustrative examples.

See Furman (2024) for a list of mathematical exercises. There is some overlap with exercises included in this tutorial, but there are also several additional exercises.

For a more in-depth introduction to the theoretical foundations of SLT, see Watanabe's two research monographs.

- Watanabe (2009) "Algebraic Geometry and Statistical Learning Theory," colloquially known as "the grey book." Derives the free energy formula in the realisable case along with other results concerning generalisation properties of Bayesian inference and maximum a posteriori inference.
- Watanabe (2018) "Mathematical Theory of Bayesian Statistics," colloquially known as "the green book." An alternative presentation of the free energy formula generalised to the non-realisable case, among other results. Compared to the grey book, the green book has an updated presentation of the main results, but does not contain all of the details of the proofs of the main results from the grey book.

While the theoretical foundation of SLT is described across many research papers by Watanabe and others, the main results are collected in self-contained form in these monographs. Watanabe offers a set of lecture slides (Watanabe, 2023) which may serve

as a useful guide to the "big picture" while working through the details in the books. Other key papers include Watanabe (2007, 2013). See Watanabe (2024) for a more detailed survey.

5.2 Recent work on singular deep learning

Over the last few years, a community of researchers have pursued the application of SLT to advancing the science and safety of deep learning. Some of the ideas behind this research are discussed by Hoogland et al. (2023); Skalse (2023); Pepin Lehalleur et al. (2025); Furman (2026). We briefly survey some key topics in this emerging literature.

Characterising and estimating degeneracy in practice. We have seen examples of parameter–function map degeneracy in simple neural networks. Symmetries of neural network parameter–function maps have long been studied; however, often the emphasis has been on discrete or globally continuous symmetries rather than additional symmetries localised to subsets of parameter space (see Farrugia-Roberts, 2022, §2.3, for a survey). For two-layer hyperbolic tangent networks, Farrugia-Roberts (2024) characterises the regions of parameter space which display additional degeneracy, and Farrugia-Roberts (2023) characterises degenerate directions in the parameter–function map.

As discussed, analytically calculating the (local) learning coefficient for deep neural networks is challenging. However, precise formulas have been derived for multi-layer deep linear networks (Aoyagi and Watanabe, 2005; Aoyagi, 2024), two-layer hyperbolic tangent networks (Aoyagi, 2009), and certain other architectures. These results assume data is generated from a known "teacher" model.

In practice, we lack knowledge of the true data generating process, and we use more complex architectures. Much work has therefore built on the foundational methods of estimating the local learning coefficient with scalable Markov chain Monte Carlo methods (Lau et al., 2025; Hitchcock and Hoogland, 2025). For a practical introduction to learning coefficient estimation, see Furman (2023). Chen and Murfet (2025) characterise the sensitivity of local learning coefficient estimation to patterns in sequence models.

Bayesian phase transitions and stagewise development. As we have discussed, Watanabe's free energy formula suggests Bayesian deep learning should undergo Bayesian phase transitions under certain conditions. Carroll (2021) studied Bayesian phase transitions in small ReLU networks, and Chen et al. (2023) studied Bayesian phase transitions in a small feature autoencoder (the "toy model of superposition" from Elhage et al., 2022).

In practice, we use stochastic gradient-based optimisation, rather than Bayesian learning, to train neural networks. However, the Bayesian case serves as a model system from which we can derive empirically testable predictions. Chen et al. (2023) formulate the *Bayesian antecedent hypothesis*, the empirical conjecture that observed phase transitions in trained neural networks correspond to Bayesian phase transitions modelled by Watanabe's free energy formula. Chen et al. (2023) study such *dynamical phase transitions* in their toy autoencoder and observe a temporal correspondence between phase changes and estimated LLC increases consistent with the free energy formula.

Wang et al. (2024); Hoogland et al. (2025) scale this methodology to transformers trained on natural language, finding similar *stagewise development* phenomena with changes in behaviour and internal structure accompanied by LLC changes. Panickssery and Vaintrob (2023); Hoogland et al. (2025); Carroll et al. (2025); Urdshals and Urdshals (2025) also study developmental stages in transformers trained on synthetic data. Elliott et al. (2026) extend this study to a case of goal misgeneralisation in deep reinforcement learning.

Degeneracy, interpretability, and patterning. The LLC provides a single number representing the effective dimensionality of a model. We can derive from the same principles—asymptotic properties of the posterior that reflect degeneracies in the model—more fine-grained tools for probing the internal computational structures of neural networks and how they depend on data.

- *Weight- and data-refined LLCs:* Wang et al. (2025b) compute LLCs of individual transformer modules (e.g., different attention heads) or with respect to different subsets of a data set (e.g., natural language versus code). Observing these refined quantities over training reveals modules developing different structures specialising to different kinds of data.
- *Susceptibilities:* Drawing inspiration from electromagnetic susceptibilities in physics Baker et al. (2025); Wang et al. (2025a); Gordon et al. (2026) develop and apply a methodology for computing loss susceptibilities so as to reveal more fine-grained information about how model internals relate to data.
- *Bayesian influence functions:* Similarly, drawing inspiration from training data attribution in statistics, Kreer et al. (2025); Lee et al. (2025); Adam et al. (2025) develop a Bayesian generalisation of classical influence functions that allows the influence functions to be sensitive to higher-order degeneracy in the model.

Weight- and data-refined LLCs are themselves LLCs with a different model or data set, and so the same scalable Markov chain Monte Carlo methods can be used to estimate them as for LLCs. Moreover, like the LLC, susceptibilities and Bayesian influence functions can be approximated as expectations over a localised posterior distribution, and so similar estimation methods can be used for these quantities too.

Wang and Murfet (2026) develop a methodology, *patterning*, for making targeted changes to the training distribution so as to elicit certain changes in the development of neural network internal structure or generalisation behaviour.

Foundations of singular deep learning. There has been some work on developing the foundational theory of SLT and deep learning. For example, Elliott et al. (2026) generalise Watanabe's free energy formula from Bayesian inference to a generalised non-stationary energy-based inference setting, so as to account for stagewise development in deep reinforcement learning. Beyond the setting of Bayesian inference, the general

role degeneracy plays in the learning dynamics of stochastic gradient-based optimisation remains to be characterised.

Finally, there has been some attempt to theoretically investigate the links between degeneracy in deep learning and computational structure in models. Clift et al. (2021); Waring (2021); Xu (2021); Murfet (2024); Murfet and Troiani (2025) study degeneracies in a statistical model based on a parameterisation of the space of Turing machine programs, exploring links between degeneracy and computational structure. Lau (2025, Chapter 5) and Urdshals et al. (2025) develop a theory of minimum description length in degenerate statistical models.

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